

Michael F. Xu

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Research Interest

My primary research interest is in the field of human-robot interaction, predominantly the social aspect of it, and integrating robotic technologies into human activities and settings. In particular, I am interested in long-term companion robot engagement and deployments, as well as their integration within family and school environments.

Education

Ph.D.	University of Wisconsin–Madison , Computer Sciences	2023 – Present
	• Advisor: Dr. Bilge Mutlu, Doctoral Minor: Human Ecology	
M.Sc.	Georgia Institute of Technology , Computer Sciences	2019 – 2021
M.A.	Yale University , Economics	2014 – 2016
B.A.	New York University , Economics, and Mathematics (<i>summa cum laude</i>)	2011 – 2014
B.A.	Tsinghua University , Economics and Finance	2010 – 2011

Experience

UW–Madison, Department of Computer Sciences , Graduate Assistant	Madison, WI
• Various research and teaching assistant roles within the department.	2023 – Present
Van Robotics , Lead Software Architect	Columbia, SC
• Full Stack Engineer (2019 – 2020).	2019 – 2023
• Design and implement the high-level software architecture of the smart robot tutor ABii; Work with research team to perform data analysis, implement machine learning models, as well as deploying the models to edge devices in real-world settings.	
Music X Lab , Research Intern	Remote
• Advised by Dr. Gus G. Xia. Implemented hardware and software solutions for haptic music tutoring systems.	2022 – 2023
IQVIA , Model Developer	New York, NY
• Develop various Excel models; significant VBA codes used both to enhance user experience, and to compute model outputs (Epidemic/Monte Carlo models).	2017 – 2018

Publications

Xu, M. F. , & Mutlu, B. (2025, March). <i>Exploring the Use of Robots for Diary Studies</i> . In 2025 20th ACM/IEEE International Conference on Human-Robot Interaction (HRI) [25%]	2025
He, X., Xu, M. F. , Cagiltay, B., & Mutlu, B. (2025, March). <i>Developing Robot Prototypes to Explore Robot-Facilitated Family Routines</i> . In 2025 20th ACM/IEEE International Conference on Human-Robot Interaction (HRI) [LBR]	
Xu, M. F. , Cagiltay, B., Michaelis, J., Sebo, S., & Mutlu, B. (2024, August). <i>Robots in family routines: Development of and initial insights from the family-robot routines inventory</i> . In 2024 33rd IEEE International Conference on Robot and Human Interactive Communication (ROMAN) [Finalist: Pioneering Research Award]	2024

Honors & Awards

Computer Sciences Summer Fellowship , University of Wisconsin-Madison	2024
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Outreach

Grand Parents University

2025

- Served as the instructor for a two-day session, teaching grandparents and grandchildren how to program robots and interact with large-language models.

UW-Madison Science Expeditions

- Hosted lab tours and stationed exhibition tables, performing multiple interactive robot demonstrations.

Lab Tour for Elementary School

- Co-hosted a half-day lab tour for a class of elementary school students, including a gentle introduction to the subject of robotics, and interactive demo sessions.

Referee Service

- IDC 2024
- CHI 2025
- HRI 2025, 2026

Personal Information

Languages: Mandarin & English

Date of Birth: Oct. 26th, 1991

Citizenship: U.S.A.